



SIMATS
ENGINEERING



SIMATS
Sreeetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Serverless Computing Reducing Infrastructure Management in the Cloud

CSA1501-Cloud Computing and Big Data Analytics

GUIDED BY:

Dr. M. D.Boomija
SIMATS ENGINEERING
CHENNAI

PRESENTED BY:

A.Reddy Prasad(192472045)
Department of CSE-AI
SIMATS ENGINEERING
CHENNAI



Definition

Serverless Computing is a cloud computing model where developers build and run applications without managing servers. The cloud provider automatically handles server management, scaling, and maintenance.

Example:

Online Food Delivery App (Swiggy / Zomato)



Introduction



- Serverless Computing is a cloud computing model.
- Developers can build and run applications without managing servers.
- The cloud provider handles server maintenance and scaling.
- It reduces infrastructure management and saves development time.
- It is widely used for web apps, mobile apps, APIs, and IoT applications.
- Popular platforms include AWS Lambda, Microsoft Azure Functions, and Google Cloud Functions.



Industry Relevance



Serverless Computing is widely used because it reduces costs and improves application performance.

Applications

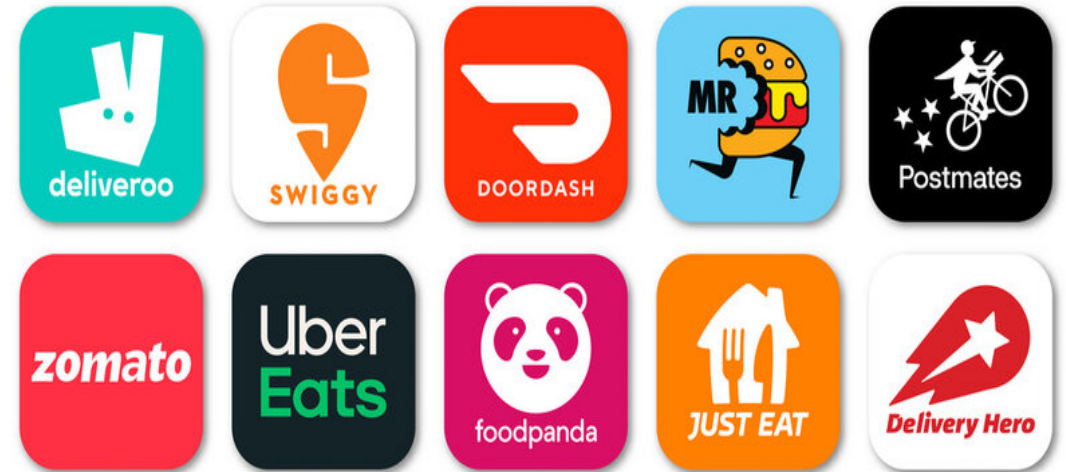
- ☐ Web Applications
- ☐ Mobile Applications
- ☐ E-commerce Websites
- ☐ Banking Systems
- ☐ Healthcare Applications
- ☐ Online Learning Platforms
- ☐ IoT Applications
- ☐ Data Processing

Practical Example

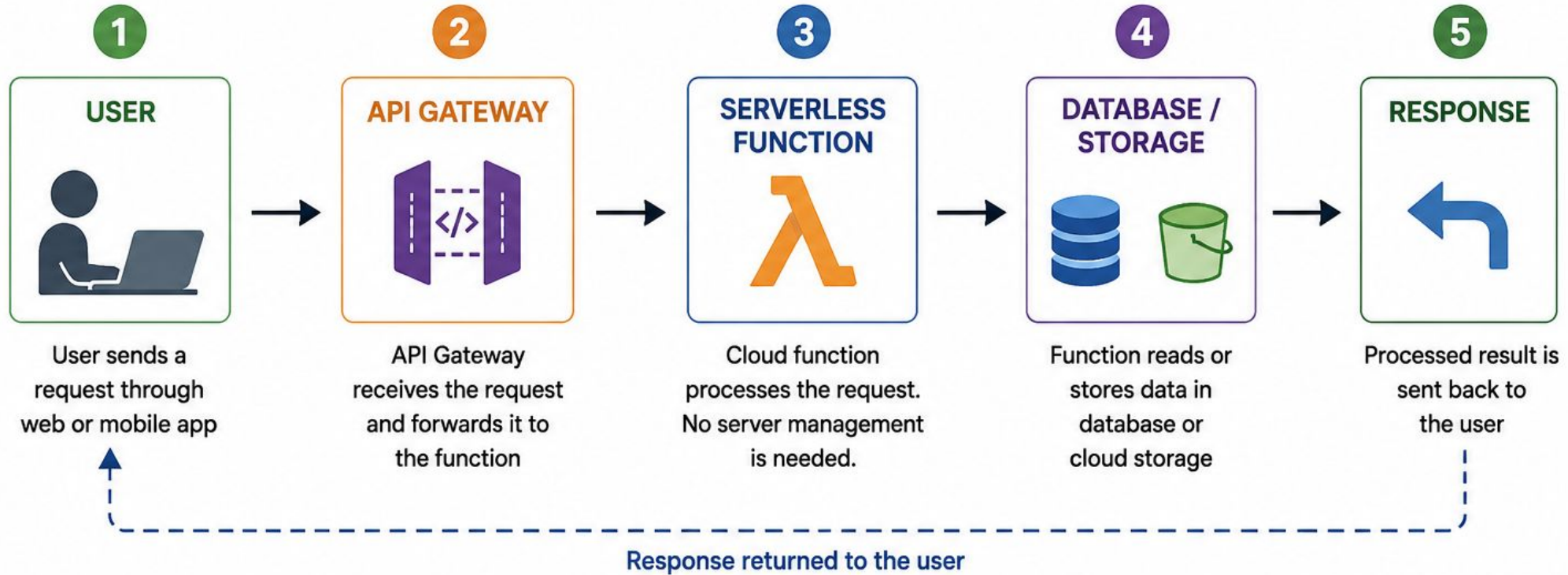
Online Food Delivery App (Swiggy / Zomato)

Process:

- ❑ User places a food order.
- ❑ API Gateway receives the request.
- ❑ AWS Lambda processes the order.
- ❑ Order details are stored in the database.
- ❑ Restaurant receives the order.
- ❑ Confirmation is sent back to the user.



Architecture diagram





Conclusion



- Serverless Computing is a modern cloud computing model that eliminates the need to manage servers.
- It reduces infrastructure costs because users pay only for the resources they use.
- It helps developers build, test, and deploy applications faster.
- Serverless computing improves performance, reliability, and flexibility.
- It is widely used in web applications, mobile apps, IoT, and cloud services.



Conclusion

A conceptual image featuring a hand in a dark suit pointing its index finger towards a glowing, circular digital button. The button has a blue gradient and a metallic, futuristic border. The text 'THANK YOU' is written in a bold, white, sans-serif font, centered on the button. The background is a dark blue space filled with a network of glowing white nodes connected by thin lines, with soft blue bokeh lights scattered throughout, creating a high-tech, digital atmosphere.

**THANK
YOU**